2026 Winter ECE 720
Group Final Project

Reliable LIME under Query Budget Constraints

Group 3

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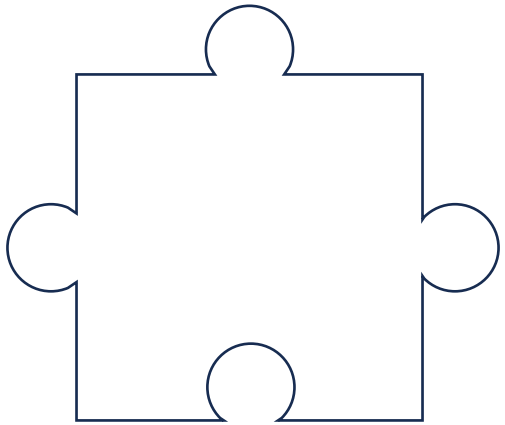
PART 1

Background and Motivation

LIME(Local Interpretable Model-agnostic Explanations)

LIME



Why did the model classify
this **specific** data as this category?

Interpretable Model

Original Model

Locality-aware weight

$$\xi(x) = \arg\min_{g \in G} \underline{L(f, g, \pi_x)} + \Omega(g)$$

$$\mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in \mathcal{Z}} \pi_x(z) \left(f(z) - g(z')\right)^2$$

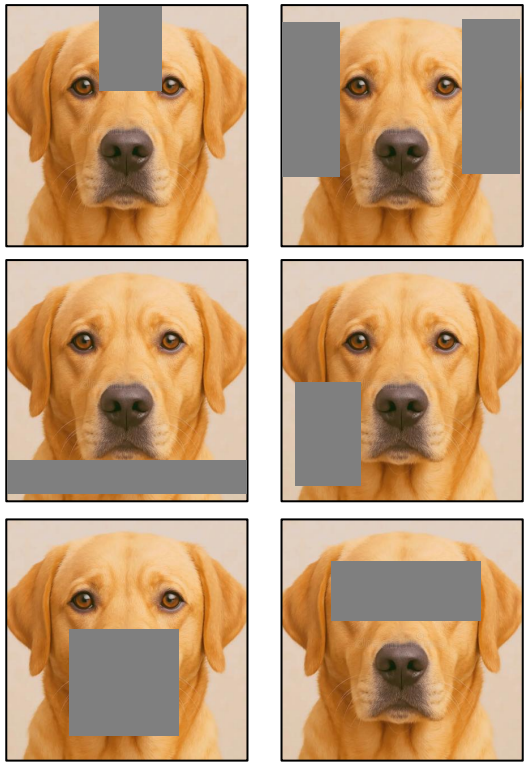
Original Input

Sample Input

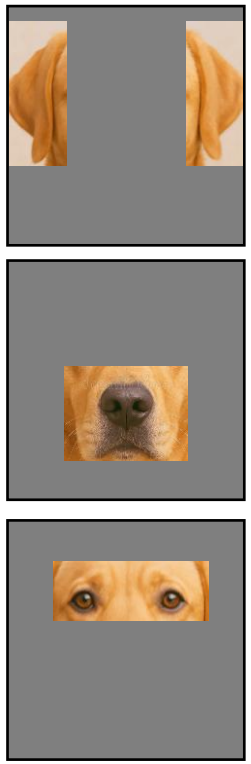
LIME(Local Interpretable Model-agnostic Explanations)

Example. Dog Classifier Model

Original Input



Sampling for Local Exploration
(Super-pixels)



Interpretable
Data Representations

Real-world Limitations

$$\xi(x) = \arg\min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

It is a time-consuming problem,
and our resources(time, money, etc.)
are not infinite.

Motivation

How can we find a more efficient explanation model

within the same budget?

Motivation

Possible Directions

Adaptive Budget	Budget Allocation Optimization		Balanced Mask Sampling
Dynamically choose the number of perturbation samples N for each instance, using more queries when the explanation is unstable and stopping early when it becomes stable.	**Coarse-to-Fine Budget Allocation** Spend a small budget to quickly find candidate important features first, then use the remaining budget to refine the explanation by focusing perturbations on those candidates.	**Repeated LIME with Split Budget and Aggregation** Split the budget N into R smaller runs, run LIME R times with different seeds, and then aggregate the feature weights (e.g., mean or median) to produce the final explanation.	Design perturbation masks so each feature is turned on and off a similar number of times, reducing sampling variance and making LIME more stable under small N

PART 2

Methodology

Numerical Metrics

We defined numerical metrics to evaluate explanations from four perspectives:

- **Fidelity**
- **Stability**
- ★ **Faithfulness**
- **Cost:** Number of model queries (budget) and runtime per explanation.

Numerical Metrics

Fidelity: Weighted Mean Squared Error (WMSE) between surrogate predictions $\hat{y}_i$ and black box predictions y_i on perturbed samples.

$$WMSE = \frac{\sum_{i=1}^{N} w_i (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} w_i}$$

Why do we need weight w_i for each surrogate explanation?

Samples closer to the original instance should matter more.

$$w_i = \pi_x(z_i),$$

z_i: the perturbation of original image x

π_x: the kernel that measures the similarity

Numerical Metrics

Stability: Jaccard overlap of top k superpixels across repeated runs with different random seeds under the same budget.

$$Stability@k = \frac{1}{\binom{m}{2}} \sum_{1 \le i \le j \le m} \frac{|T_i \cap T_j|}{|T_i \cup T_j|}$$

Where:

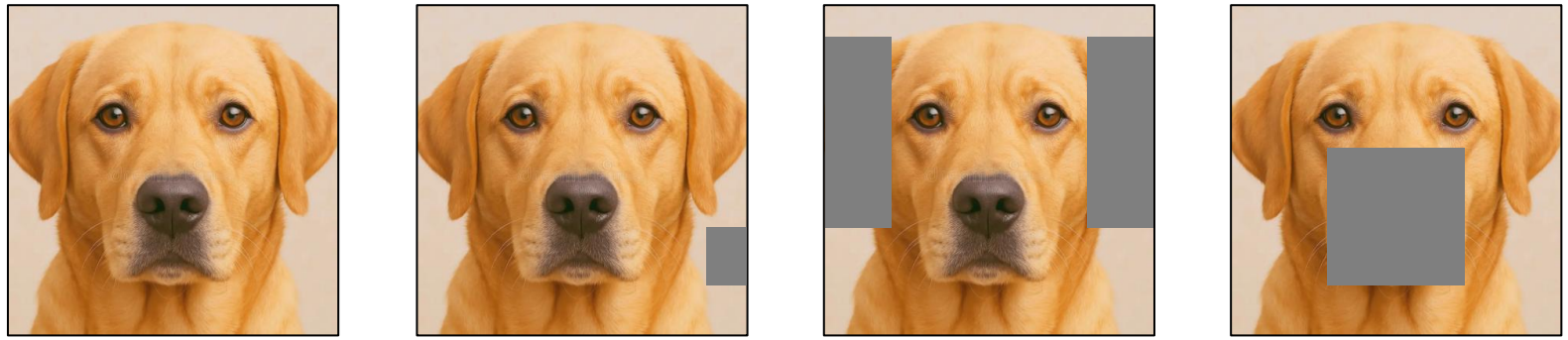
m: the number of runs

T_i, T_j: the selected top-k superpixels in the corresponding round

Numerical Metrics

★ **Faithfulness:** Drop of black box prediction probability after removing the top-k superpixels which is selected by the explainer.

$$FaithDrop@k = f(x) - f(x_{T_k})$$



Original Input	Deletion Test1	Deletion Test2	Deletion Test3
Dog: 0.973	Dog: 0.972	Dog: 0.908	Dog: 0.857
	FaithDrop≈0	**FaithDrop=0.065**	**FaithDrop=0.116**

- Larger drop → the selected superpixels are more influential
- Smaller drop → the selected superpixels are less influential

Budget Sweep Pipeline

Algorithm 1 Budget Sweep Evaluation with Metrics

Require: Explainer g, Image x, Segmentation S, target label c
Require: Budge list $budgets$, Seed list $seeds$, Explanation length K

$M \leftarrow \{\}$
for $budget \in budgets$ **do**
  $E \leftarrow \{\}$
  for $seed \in seeds$ **do**
    $e \leftarrow g(x, S, c, budget, seed)$
    $T \leftarrow \text{top_k_superpixels}(e, K)$
    $faith, fid \leftarrow \text{FaithDrop}(x, S, T, c), WMSE(e, c)$
    $E \leftarrow E \cup < e, faith, fid >$
  end for
  $M \leftarrow M \cup < budget, \text{StabilityJaccard}(E, K), E >$
end for
return M

Coarse-to-Fine Budget Allocation

Algorithm 2 Coarse-to-Fine Budget Allocation

Require: Classifier f, Budget N
Require: Image x, its interpretable version x'
Require: Similarity kernel π_x, Explanation length K
Require: Candidate size C, Exploration ratio ε

$\alpha \leftarrow \text{adaptive_coarse_ration}(N)$
$N_1, N_2 \leftarrow \alpha N, (1-\alpha)N$
Stage 1: Coarse
$Z_1 \leftarrow \text{Coarse}(f, x', \pi_x, N_1)$
$wcoarse \leftarrow \text{SparseLinearModel}(Z_1, \max(C, K))$
$Cset \leftarrow \text{get_superpixels}(wcoarse)$
Stage 2: Fine
$Z_2 \leftarrow \text{Fine}(f, x', \pi_x, N_2, Cset, \varepsilon)$
$Z \leftarrow Z_1 \cup Z_2$
$w \leftarrow \text{SparseLinearModel}(Z, K)$
return w

Coarse-to-Fine Budget Allocation

Algorithm 2 Coarse-to-Fine Budget Allocation

Require: Classifier f, Budget N
Require: Image x, its interpretable version x'
Require: Similarity kernel π_x, Explanation length K
Require: Candidate size C, Exploration ratio ε

Stage 1: Coarse
$Z_1 \leftarrow \{\,\}$
for $i \in \{1, 2, 3, \ldots, N_1\}$ **do**
  $z_i' \leftarrow \text{sample_around}(x')$
  $Z_1 \leftarrow Z_1 \cup < z_i', f(z_i), \pi_x(z_i) >$
end for

Coarse-to-Fine Budget Allocation

Algorithm 2 Coarse-to-Fine Budget Allocation

Require: Classifier f, Budget N
Require: Image x, its interpretable version x'
Require: Similarity kernel π_x, Explanation length K
Require: Candidate size C, Exploration ratio ε

```
# Stage 2: Fine
Z_2 ← { }
for j ∈ {1, 2, 3, ..., N_2} do
    if rand() < ε then
        z'_j ← sample_around(x')
    else
        z'_j ← sample_focused_on(Cset)
    end if
    Z_2 ← Z_2 ∪ < z'_j, f(z_j), π_x(z_j) >
end for
```

PART 3

Experiments, Evaluation and Conclusion

Experiment Setup

General Settings

- **Black-box model:** ResNet-18 (pretrained, ImageNet)
- **Test image:** skimage.data.cat() → class 285 “Egyptian cat”
- **Segmentation:** SLIC, 100 superpixels, fixed across all runs
- **Budget levels:** N ∈ {200, 500, 1000, 2000, 3000, 5000}
- **Repetitions:** 10 random seeds per budget
- **Surrogate:** Ridge regression (α=1), cosine distance, exp kernel (width=0.25)
- **Top-k = 10,** hide_color = 0 (black)

Coarse-to-Fine Settings

- **Adaptive coarse fraction:**
  - 50% when N ≤ 500
  - 30% when N ≤ 2000
  - 20% otherwise
- **candidate_k = 20**
- **ε = 0.2**
- **batch_size = 32**

Baseline Results: Vanilla LIME

Budget (N)	Stability (Jaccard)	Fidelity (WMSE)	Faithfulness (Drop-k)	Runtime (s)
200	0.063	0.000770	−0.056	2.74
500	0.079	0.000549	−0.019	6.80
1000	0.086	0.000343	0.023	13.85
2000	0.183	0.000267	0.155	27.93
3000	0.272	0.000223	0.145	42.10
5000	0.348	0.000163	0.193	69.71

Key Findings:

- At N ≤ 1000: Faithfulness is negative → explanation points to wrong regions
- Stability < 0.1 → explanations are not reproducible

Baseline Results: Visual Explanation Quality

LIME, N=200
285 Egyptian cat

LIME, N=500
285 Egyptian cat

LIME, N=1000
285 Egyptian cat

LIME, N=2000
285 Egyptian cat

LIME, N=3000
285 Egyptian cat

LIME, N=5000
285 Egyptian cat

Baseline Results: Visual Explanation Quality

LIME, N=8000
285 Egyptian cat

LIME, N=12000
285 Egyptian cat

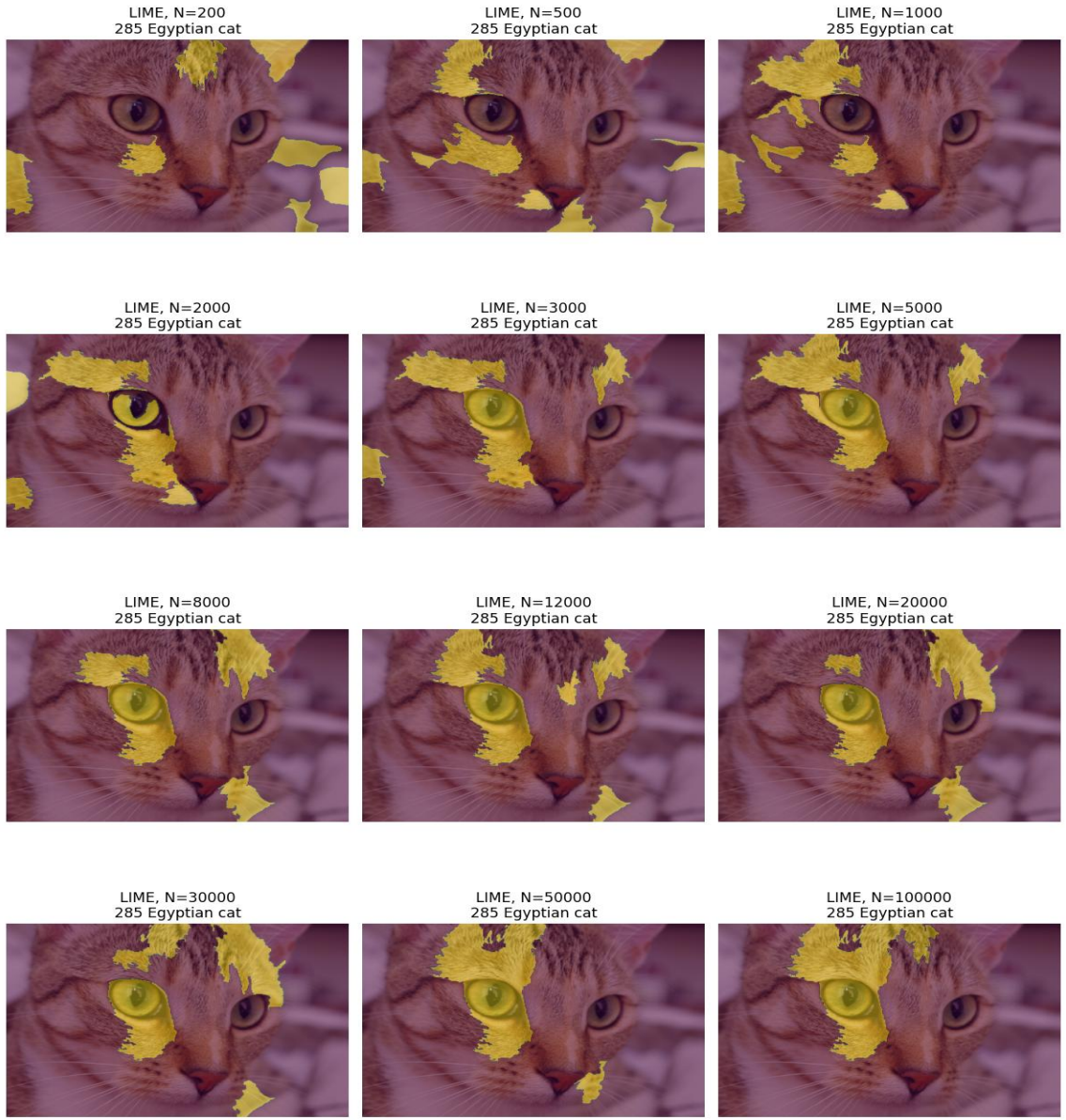
LIME, N=20000
285 Egyptian cat

LIME, N=30000
285 Egyptian cat

LIME, N=50000
285 Egyptian cat

LIME, N=100000
285 Egyptian cat

Baseline Results: Visual Explanation Quality



- **N=200:** Scattered, inconsistent highlights — explanation is noise
- **N=5000:** Consistently focuses on cat's facial features (eyes, nose)

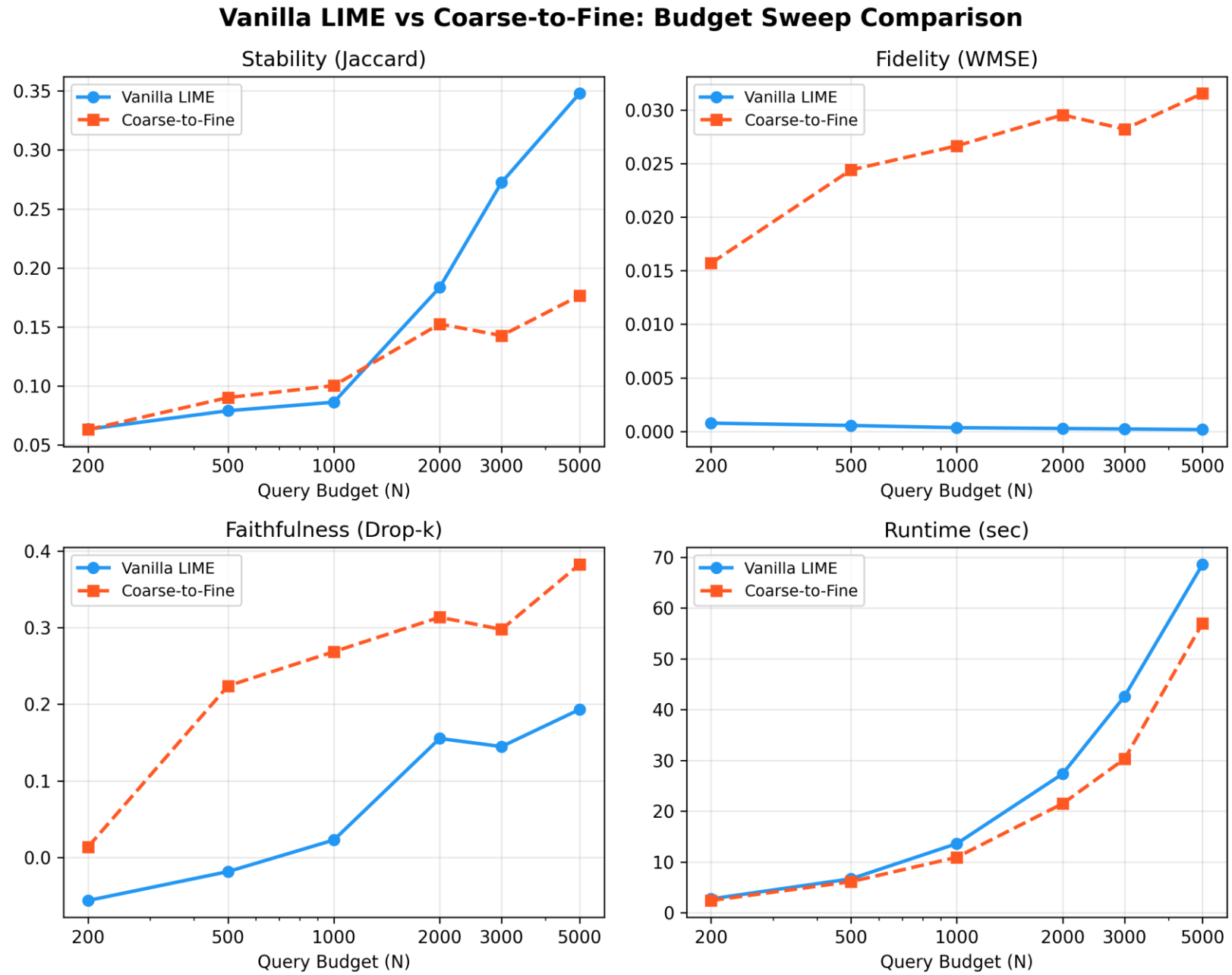
Improvement Results: Coarse-to-Fine

Budget (N)	Stability (Jaccard)	Fidelity (WMSE)	Faithfulness (Drop-k)	Runtime (s)
200	0.063	0.01571	0.014	2.40
500	0.090	0.02442	0.224	6.10
1000	0.100	0.02667	0.269	10.92
2000	0.153	0.02956	0.314	21.48
3000	0.143	0.02823	0.298	30.29
5000	0.177	0.03156	0.383	57.01

Key Findings:

- Faithfulness positive at ALL budgets (vanilla is negative at $N \leq 500$)
- At N=500: Drop-k improves from −0.019 → 0.224

Vanilla LIME vs. Coarse-to-Fine: Metric Comparison



Per-Metric Summary:

Faithfulness

C2F consistently higher

Fidelity (WMSE)

C2F much higher — focuses on harder regions

Stability

Mixed — slightly better at low N, worse at high N

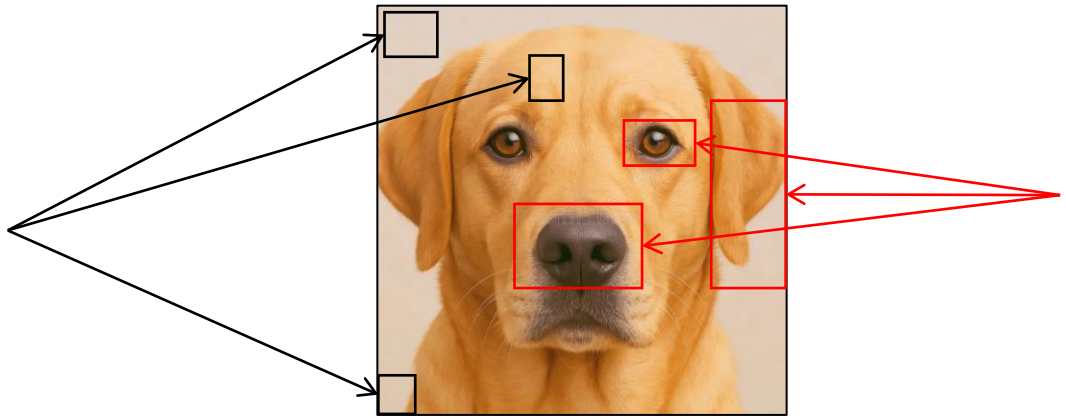
Runtime

Comparable across both methods

Why fidelity become worse than original?

Neutral area

Easy to approximate
(near-zero)

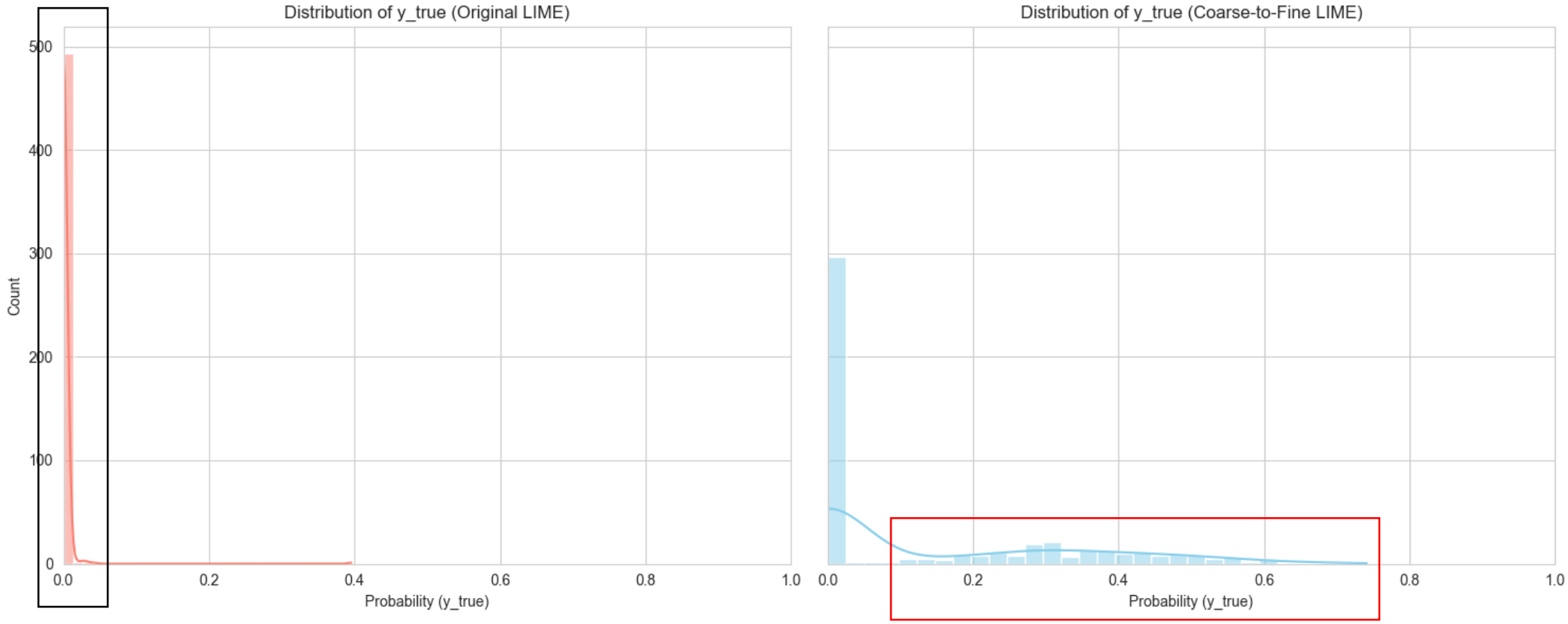


Influential area

Critical to distinguish
(fine-grained)

Why fidelity become worse than original?

Coarse-to-Fine LIME focuses on more sensitive area



(N=500)

Conclusion & Future Work

Conclusion

- Built budget sweep evaluation framework with 4 metrics
- Vanilla LIME is unreliable when N < 1000
- Coarse-to-fine improves faithfulness at all budgets, highlights meaningful regions at N=500
- No additional query overhead

Future Work

- Test on more images and models (VGG, ViT)
- Explore alternative sampling strategies for better stability
- Automatic hyperparameter tuning (coarse fraction, ε, candidate_k)
- Extend to tabular/text LIME

Thank you!